Junho Kim

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Personal
LinkedIn

Research Interests

Computational Imaging, Optics, Computer Vision, Prompt Engineering, Robotics

Education

- **Rice University** *Expected May 2026*
  B.A. in Computer Science | B.A. in Electrical Engineering GPA 3.4
- **Coursework**: Algorithms & Data Structures, Object-Oriented Prog, Machine Learning, Computer Systems, 3D Vision, Sensor Data, Imaging Optics, Computer Ethics, Linear Algebra, Statistics & Probability, Multivariable Calc

Research Experience

- **Kavraki Lab** *October 2023 – Present*
  Research Assistant
  - Leveraged dataclass decorator for type-hinted field customization in designing a robot manipulation API
  - Conducted prompt experimentation to optimize language-guided robot skill acquisition pipeline, refining code generation using Language Models (LLM) like Code Llama, GPT4
- **Rice Computational Imaging Lab** *August 2023 – Present*
  Research Assistant
  - Developed a gradient descent optimization pipeline for image reconstruction integrated with lensless cameras
  - Revised the evaluation code for the Face Detection Data in C++ to enhance compatibility with detection networks
  - Programmed an interactive data collection interface to adjust light exposure for face images
- **Tesser** *May 2023 – August 2023*
  Research Intern
  - Architected a two part hybrid network consisting both convolutional-based and attention-based mechanism, achieving a 96% accuracy in classifying histopathological breast cancer images
  - Enhanced the data preprocessing pipeline for histopathological slides by implementing bitwise contour methodologies, yielding a 12% enhancement in Dice score
  - Developed a user-friendly GUI with PyQt5 for breast cancer classification, deployed for clinical use. Achieved a 66% cost reduction in diagnoses and 3% processing time reduction through multi-threading program design
- **Center for Axion and Precision Physics, Institute for Basic Science** *June 2019 – September 2019*
  Research Intern
  - Confirmed the Planck constant through induced Shapiro steps from Superconducting Quantum Interference device experimentation
  - Experimented with varies working points on the V-Φ curve to find the optimal balance of noise amplification

Technical Projects

- **Rice Robotics Club | Camera Vision Engineer** *January 2024 - Present*
  - Adapted ROS movement code for cat leg motion
  - Implemented a monocular vision pipeline for robot navigation
  - Developed a prototype for a cat leg using Fusion360 and 3D printing
- **Rice Datathon (Chevron Challenge) | Team Lead** *January 2024*
  - Minimized RMSE for oil production prediction through diverse ML methods: K-means Clustering, XGBoost, Neural Networks, and Lasso/Ridge Regression
  - Performed categorical and numerical analysis on diverse data types to calculate parameter correlations
- **Rice Datathon (Energy Track) | Software Developer** *January 2023*
  - Conducted principal component analysis of 32 environmental factors to visualize and gain insight in return on investment for renewable energy projects per state
  - Applied linear, polynomial, and LASSO regression models to minimize root mean squared error
- **Open Dataset Classification** *September 2022*
  - Streamlined and automated the weight initialization process for weights and activation functions
  - Conducted extensive experiments using VGG, ResNet, and InceptionNet on FashionMNIST, CIFAR-10 datasets

Skills

Programming Languages: Python (Professional), Java (Intermediate), C (Intermediate), C++ (Novice), Javascript (Novice), SQL (Novice)

Tools/Libraries: PyTorch, OpenCV, Open3D, Weight & Biases, Docker, Github, Scikit-learn, Pandas, Flask, PyQt5, Fusion360, ROS2

Languages: English (Fluent), Korean (Native)

Professional Experience

- **Republic of Korea Army** *June 2021 – December 2022*

Radio Transmission Specialist, Squad Leader

  - Skillfully operated the mobile command tower to ensure maintenance and optimal functioning of the radio network
  - Constructed a Microsoft Excel spreadsheet that automated the registration process for wireless radio transmitters
  - Consulted over 15 military personnel, addressing topics such as military adaptation, interpersonal relationships, seniority concerns, and suicide prevention

- **MyCelebs Inc** *February 2020 – June 2020*

Data Analyst Intern

  - Created a text classifier through Python string parsing that proficiently identifies muted letters (i.e. k, p, h, r)
  - Leveraged the mute letter detector to interpret voice-reconstructed outputs into complete sentences, resulting in a 13% enhancement in sentence construction accuracy

Teaching Experience

- **Teaching Assistant,** ELEC 220 - Fundamentals of Computer Engineering *August 2023 - Present*
- **Teaching Assistant,** KORE 264 - Korean II *January 2024 - Present*
- **Teaching Assistant,** KORE 141 - First Year Korean I *August 2023 - December 2023*

Awards / Leadership

- **Scholar,** Korea Presidential Science Scholarship *2020 - Present*
- **School Representative,** Korean-American Scientists and Engineers Association *2024 - Present*
- **Camera Vision Team Lead,** Rice Robotics Club *2024 - Present*